AI 释能：感知、计算、连接

Marco Chung

2025 年 3 月 14 日

german brand award '24 corporate brand of the year

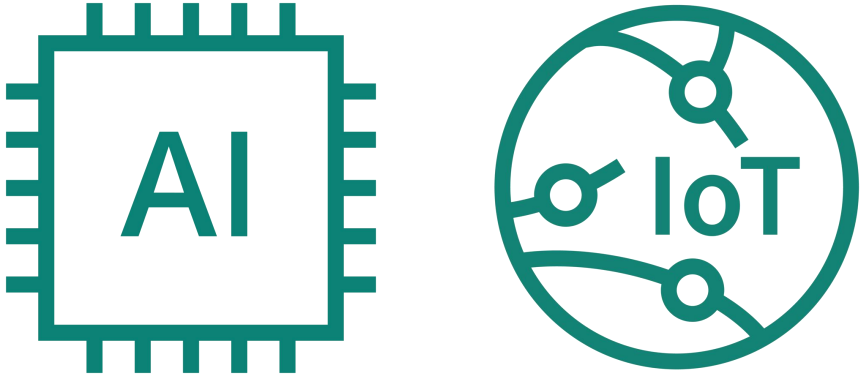
restricted

AIoT 的基础

人工智能

能够分析数据、从模式中学习并无需人类明确干预即可做出决策的智能系统

AIoT



物联网

由嵌入传感器、软件和其他技术的物理对象（“物”）组成的网络，使这些设备能够通过互联网连接并交换数据

智能家居

智慧医疗

智慧农业

智能汽车

工业物联网

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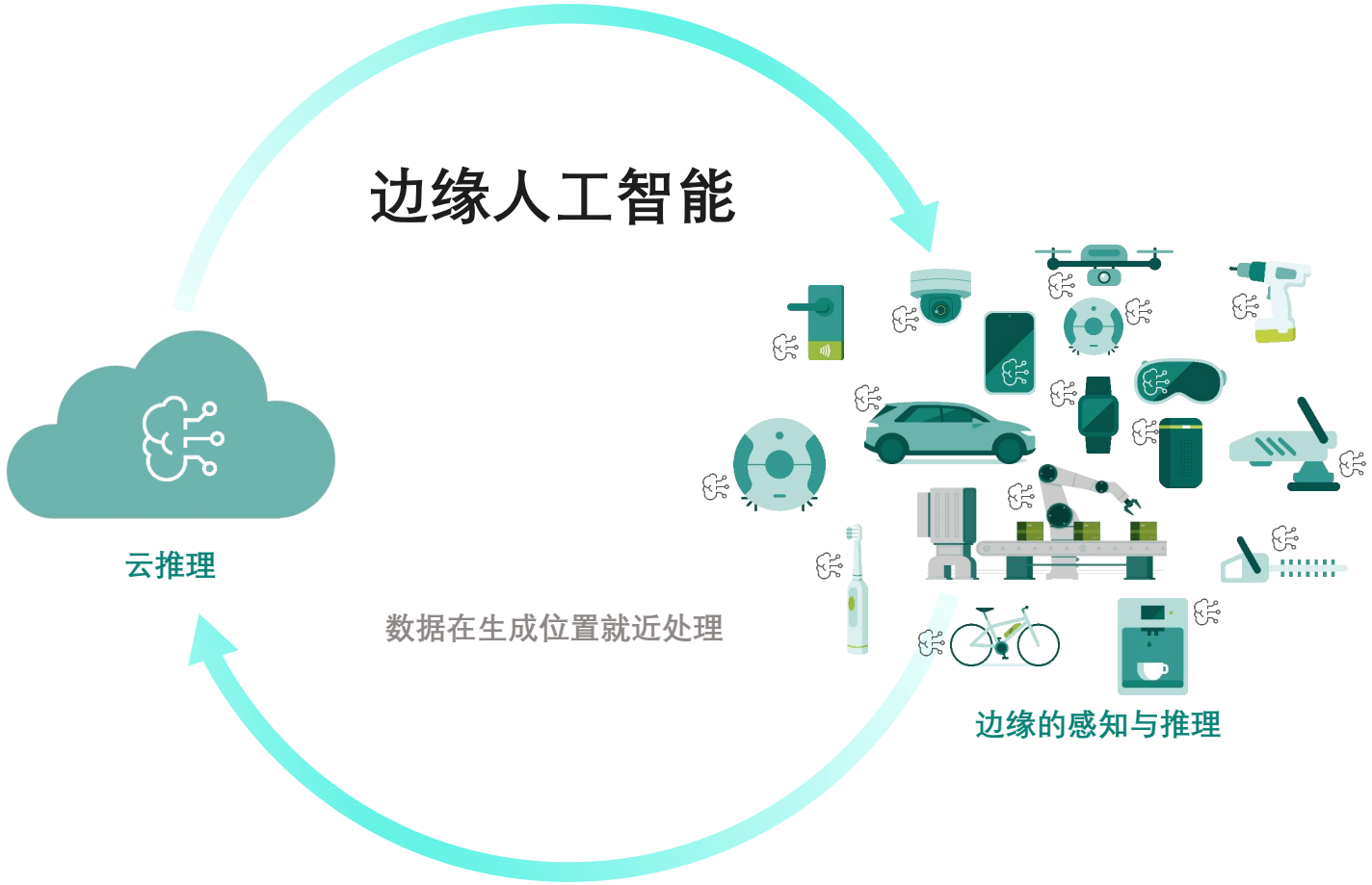
实时性需求以及对能效、安全性和隐私的重视，推动了人工智能在边缘计算中的处理发展



边缘人工智能的主要优势

- 无需互联网连接
- 低延迟和实时响应
- 优化能效
- 提高安全性和数据私密性
- 降低成本

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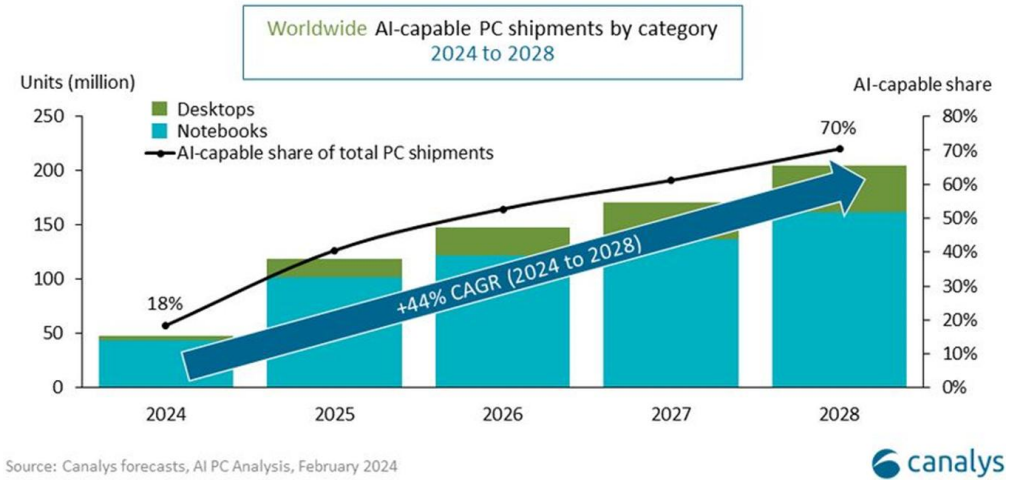
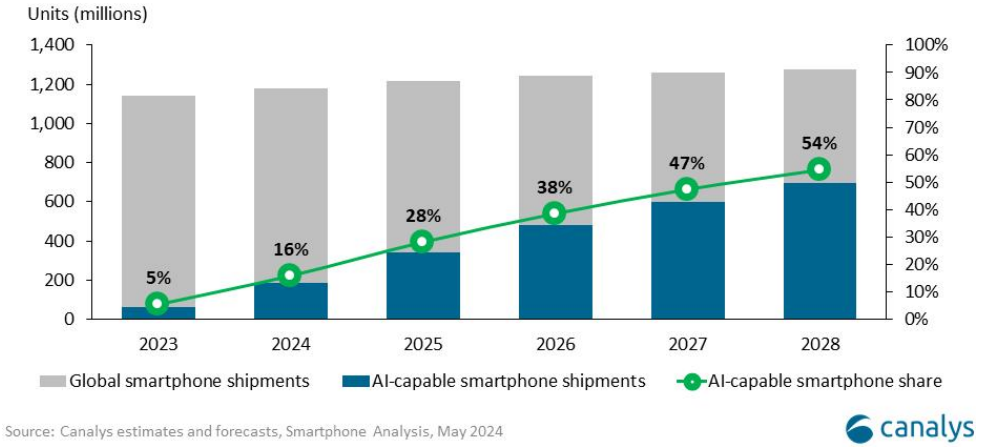
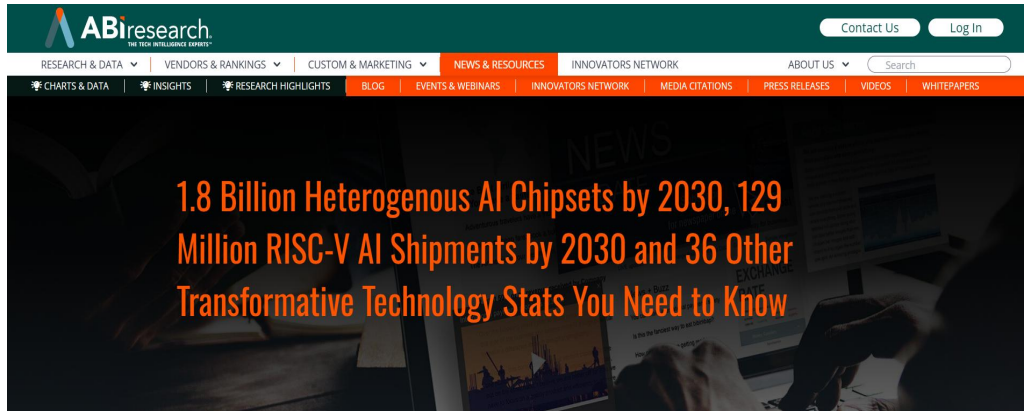
市场增长趋势

150 million+ AI-capable PCs to ship by end of 2025



AI-capable smartphone share to reach 54% in 2028





Shipments of heterogenous AI chipsets at the edge will reach more than 1.8 billion by 2030.

Robot Unit Number (Millions) Forecast by Type Source: Citi GPS

	2024	2025	2030	2035	2050	CAGR
Autonomous Vehicle	27	34	126	380	1,858	17.4%
Domestic Cleaning Robot	286	326	541	793	1,188	5.3%
Humanoids	0	0	1	13	648	60.7%
AGV & AMR	2	3	9	28	181	17.7%
Drone	37	40	54	73	149	5.4%
Caring Robot	0	1	6	18	71	20.0%
Commercial Cleaning Robot	1	2	6	14	25	10.6%
Food & Grocery Delivery Robot	0	0	4	11	19	19.2%
Food Service Robot	0	0	3	10	15	17.2%
Caring Overlap	0	0	(1)	(5)	(18)	
Total	354	405	749	1,337	4,136	9.7%

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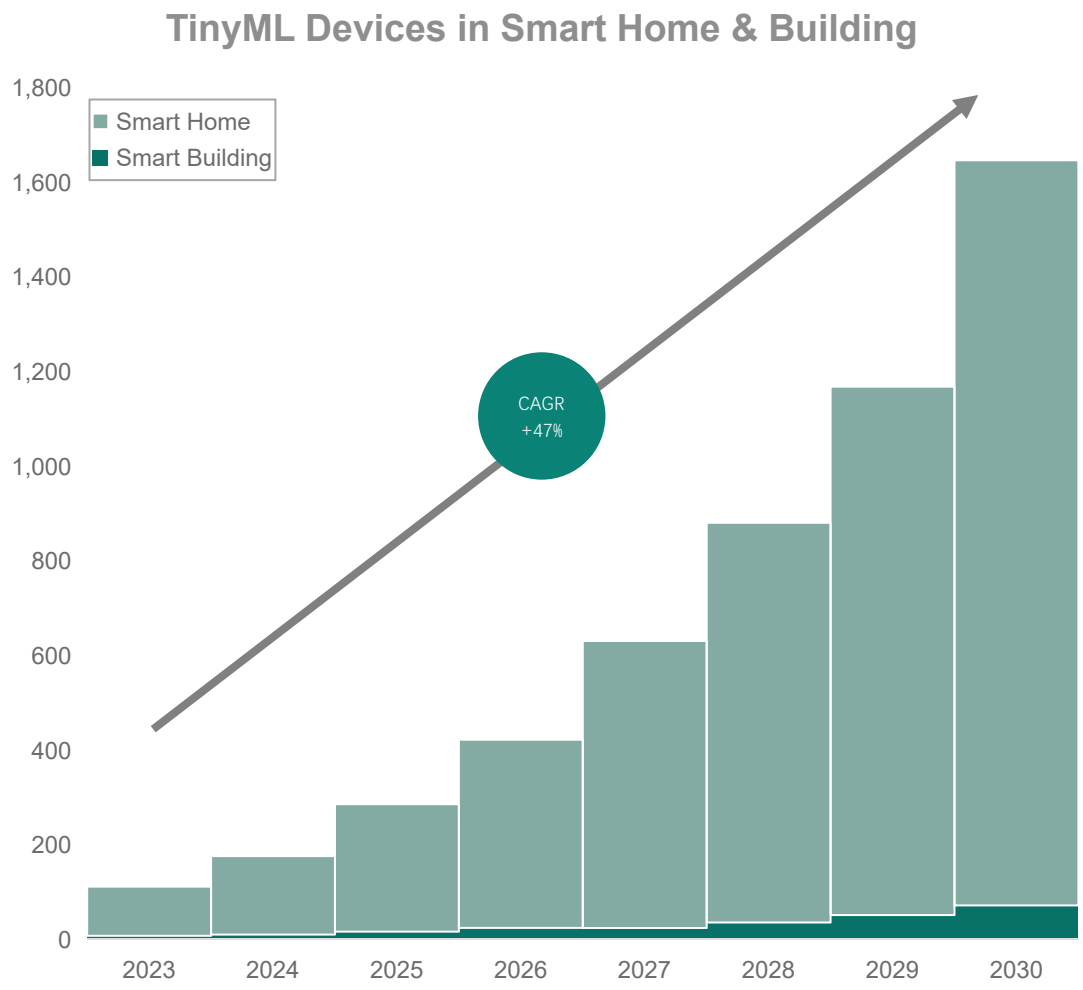
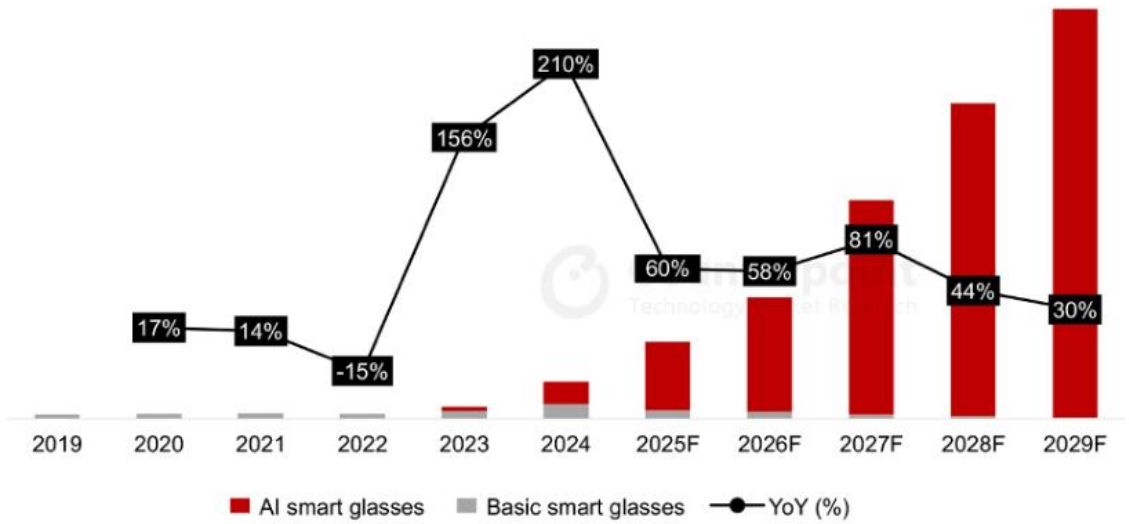
市场增长趋势



Total Shipments for TinyML Devices (Millions)	2024	2025	2026	2027	2028	2029	2030	CAGR 24-30
Agriculture	0.5	0.7	1.0	1.2	1.2	1.8	2.9	34%
Automotive	46.2	72.6	113.6	176.2	271.9	416.5	631.0	55%
Banking and Finance	0.0	0.0	0.0	0.0	0.0	0.0	0.0	75%
Cellular Network	9.9	14.5	19.8	25.3	29.6	35.2	39.5	26%
Healthcare	1.4	2.0	3.1	4.3	4.9	5.4	5.6	26%
Manufacturing	7.6	14.5	27.0	35.6	52.0	77.7	111.1	56%
Personal and Work Devices	1,036.4	1,355.2	1,608.6	1,879.9	2,201.0	2,626.6	3,312.8	21%
Retail	1.3	1.5	2.4	3.0	3.7	4.5	5.3	27%
Robotics	2.1	3.3	5.2	8.2	13.0	20.5	32.5	58%
Small UAS	0.1	0.1	0.1	0.2	0.4	0.6	0.9	61%
Smart Building	10.1	16.4	24.3	24.1	35.7	51.4	71.8	39%
Smart City	24.4	33.5	47.0	49.3	48.2	49.4	58.4	16%
Smart Home	165.8	269.2	397.6	606.5	844.7	1,116.6	1,574.6	46%
Transport & Logistics	42.2	51.7	64.0	77.2	100.7	128.7	166.0	26%
Utilities	32.1	43.2	57.0	65.6	64.4	60.2	68.0	13%
Wearables	7.8	9.7	13.2	20.5	26.4	33.2	40.7	32%
XR (head worn devices, smart glasses)	1.0	2.5	5.6	11.8	22.2	37.3	57.2	96%
Total	1,388.7	1,890.4	2,389.4	2,989.1	3,719.9	4,665.6	6,178.2	28%

Global Smart Glasses Market Shipments Volume and YoY Growth (%), 2019-2029F



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我们赋能人工智能

感知

精准可靠地感知

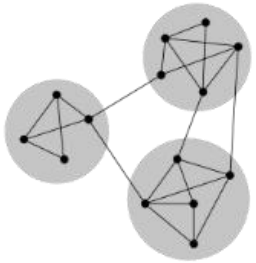
XENSIV™ 智能传感器
在汽车、工业和消费电子市场中
感知世界

计算

处理原始数据

PSOC™ Edge
支持机器学习的
下一代 MCU

连接



安全连接

AIROC™ 无线解决方案
为物联网应用提供稳健、安全的
连接性

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未来展望

无缝自动化和高效率：
无需人工干预的全自动系统，可优化资源利用率并简化流程

数字化 ：
将数字技术融入生活的方方面面，从智能家居到互联城市

低碳化：
通过优化能源消耗、减少浪费和实现高效的资源管理，成为全球可持续发展的核心

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